

Areas and Volumes — Problem Set

Warm-up

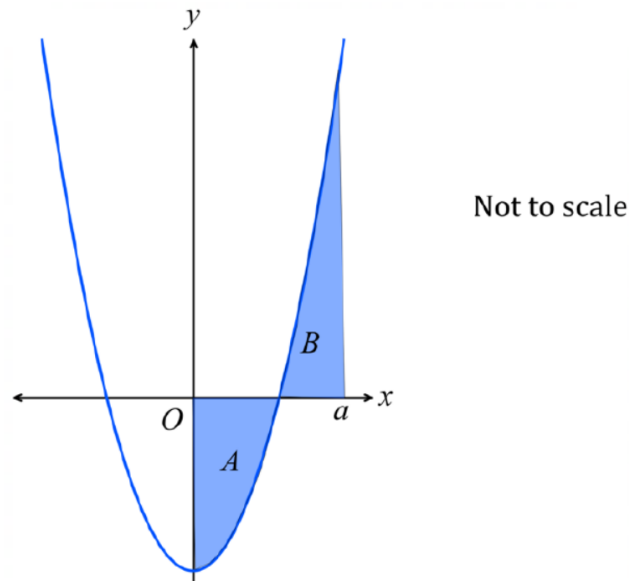
1. Find the area bounded by the curve $y = x^2 - 9$ and the x -axis.
2. Calculate the volume of the solid formed when the line $y = 2x$ from $x = 0$ to $x = 3$ is rotated about the x -axis.
3. Find the value of $\int_0^1 \frac{1}{x^2+1} dx$.
4. Determine the area of the region bounded by $y = e^x$, the x -axis, and the lines $x = 0$ and $x = 2$.
5. Find the volume of the solid generated when $y = \sqrt{x}$ between $x = 0$ and $x = 4$ is rotated about the x -axis.
6. Find the area bounded by $y = \frac{1}{x}$, the x -axis, and the lines $x = 1$ and $x = e$.
7. Calculate the volume of the solid formed by rotating the region bounded by $y = 3$, the y -axis, and $y = x$ about the y -axis.

Skill-building

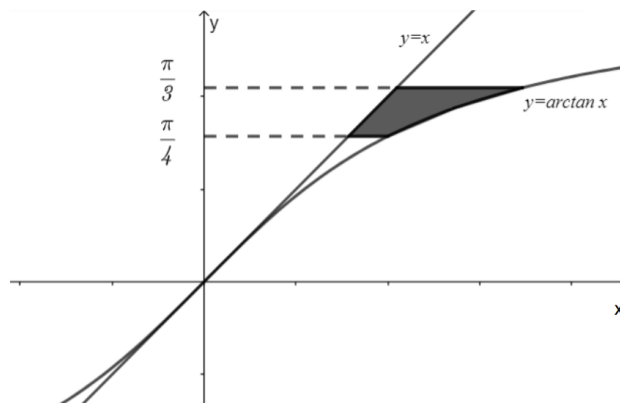
1. The region bounded by $y = \cos x$, the x -axis, and the lines $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x -axis. Find the exact volume.
2. Find the area of the region bounded by the curve $y = \tan^{-1} x$, the y -axis, and the line $y = \frac{\pi}{4}$.
3. A solid is formed by rotating the region bounded by $y = e^{2x}$, $y = 1$, and $x = 2$ about the x -axis. Calculate the volume in terms of e .
4. Find the area enclosed between the parabola $y = 4 - x^2$ and the line $y = x + 2$.
5. Use the identity $\sin^2(ax) = \frac{1}{2}(1 - \cos(2ax))$ to find the volume generated when $y = 4 \sin(3x)$ from $x = 0$ to $x = \frac{\pi}{6}$ is rotated about the x -axis.
6. The region bounded by $y = \frac{1}{\sqrt{x^2+1}}$, the x -axis, and the lines $x = 0$ and $x = 1$ is rotated about the x -axis. Find the volume of the solid.
7. Find the exact area of the region in the first quadrant bounded by $y = \ln x$, the x -axis, and the line $x = e^2$.
8. The region between $y = x^2$ and $y = \sqrt{x}$ is rotated about the x -axis. Calculate the volume of the resulting solid.
9. Consider the function $y = 2 \cos^{-1}(x/2)$. Find the area bounded by the curve, the x -axis, and the y -axis.
10. Find the volume formed when the region bounded by $x = y^2 + 2$, the y -axis, and the lines $y = 0$ and $y = 2$ is rotated about the y -axis.

Easier Exam Questions

1. (Hunters Hill 2023 Q9) The graph of $y = x^2 - 4$ is shown below. The area of the region A is equal to the area of the region B. What is the value of a ?



- (A) 6
 (B) $\sqrt{6}$
 (C) $2\sqrt{3}$
 (D) 12
2. (Fort Street 2022 Q15b) Find the volume formed when the region between the curve $y = \tan^{-1} x$ and the lines $y = x$, $y = \frac{\pi}{3}$ and $y = \frac{\pi}{4}$, is rotated around the y -axis. 4



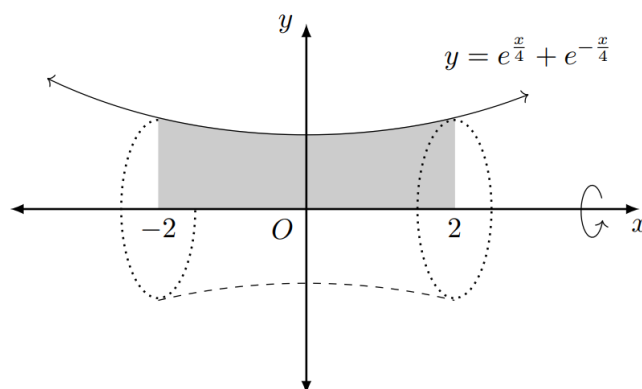
3. (Cherrybrook Tech 2021 Q13c) A solid is generated when the region bounded by $y = \frac{1}{x}$, $y = 4$, the line $x = 3$ and the y -axis is rotated about the x -axis. 3

Determine the volume of the solid.

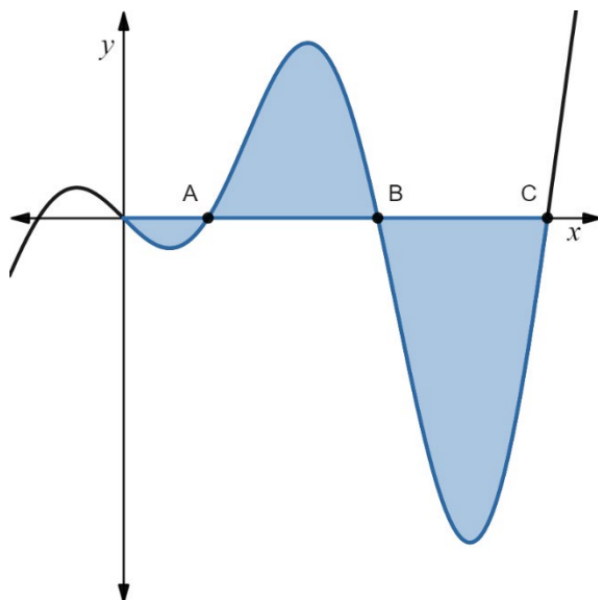
4. (Ascham 2024 Q11f) To make a solid rim for a wheel, the region between the curve $y = e^{\frac{x}{4}} + e^{-\frac{x}{4}}$, the lines $x = -2$, $x = 2$ and the x -axis is rotated around the x -axis as shown below.

Find the volume of the solid formed.

3



5. (Sydney Girls 2022 Q14c) The diagram below represents the graph of the function $y = \sin 5x - \sin 6x$.
- (i) Determine the exact coordinates of the point C . **2**
- (ii) Hence, calculate the volume of the solid formed when the shaded region is rotated by 360° about the x -axis. Give your answer correct to 2 decimal places. **3**



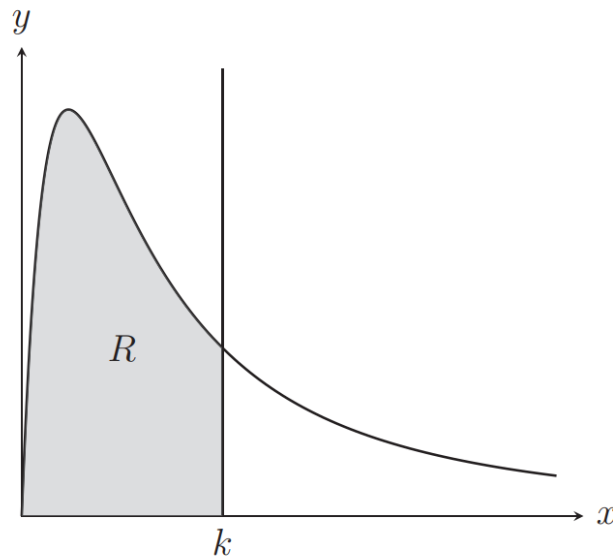
Harder Exam Questions

1. (Sydney Grammar 2025 Q13b) The diagram below shows the graph of the function $f(x) = \frac{\pi}{\pi x + 1} \sin\left(\frac{\pi}{\pi x + 1}\right)$ for $x \geq 0$. The shaded region R is bounded by the graph of $y = f(x)$, the x -axis, and the vertical line $x = k$, where $k > 0$.

- (i) Show that the volume V of the solid generated when R is rotated about the x -axis can be found by evaluating the integral: **2**

$$V = \frac{\pi^3}{2} \int_0^k \frac{1}{(\pi x + 1)^2} \left[1 - \cos\left(\frac{2\pi}{\pi x + 1}\right) \right] dx.$$

- (ii) Use the substitution $u = \frac{2\pi}{\pi x + 1}$ to show that $V = \frac{\pi}{4} \left[\sin\left(\frac{2\pi}{\pi k + 1}\right) + \frac{2\pi^2 k}{\pi k + 1} \right]$. **3**



(iii) What does the volume V approach as $k \rightarrow \infty$?

1

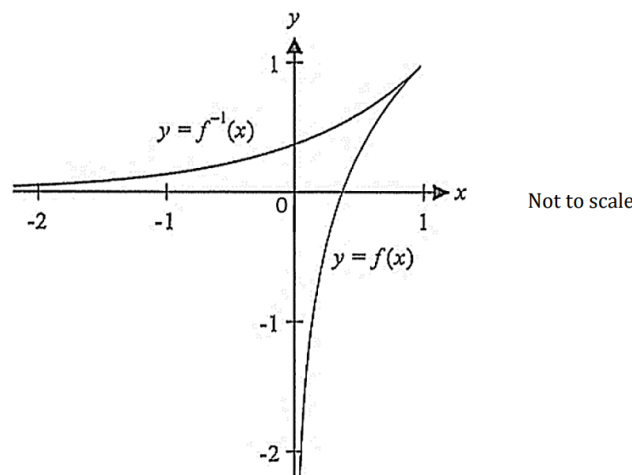
2. (St George Girls 2020 Q13c) The function $f(x) = 1 + \ln x$ is defined in the domain $(0, 1]$.

(i) Show that $\frac{d}{dx}(x \ln x) = 1 + \ln x$.

1

(ii) The diagram shows the graphs of the function $y = f(x)$ and the inverse function $y = f^{-1}(x)$. Find in simplest exact form the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = f^{-1}(x)$ and the coordinate axes.

3



3. (Riverview 2024 Q13d) Let $f(x) = \ln\left(\frac{2x}{3}\right)$.

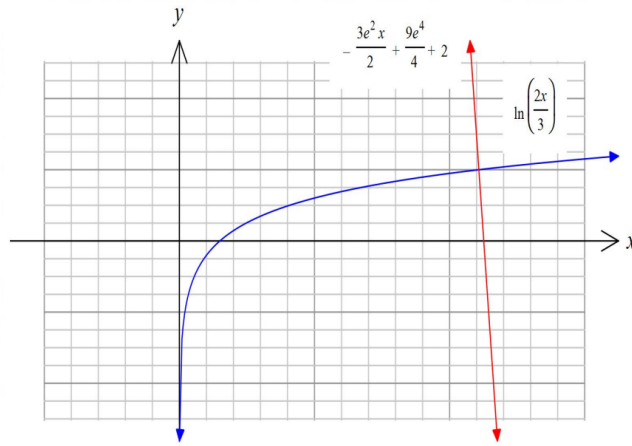
(i) Show that the equation of the normal to the graph of $y = f(x)$ at the point where $x = \frac{3e^2}{2}$ has the equation $y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$.

2

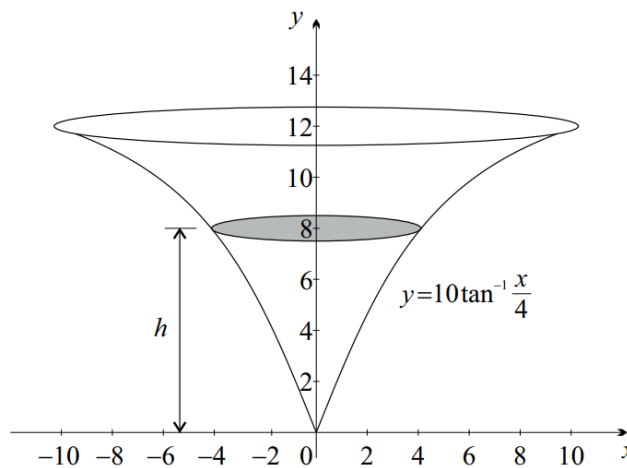
(ii) Find the exact area enclosed by the graph of $y = f(x)$, the line $y = -\frac{3e^2x}{2} + \frac{9e^4}{4} + 2$ and the x axis.

3

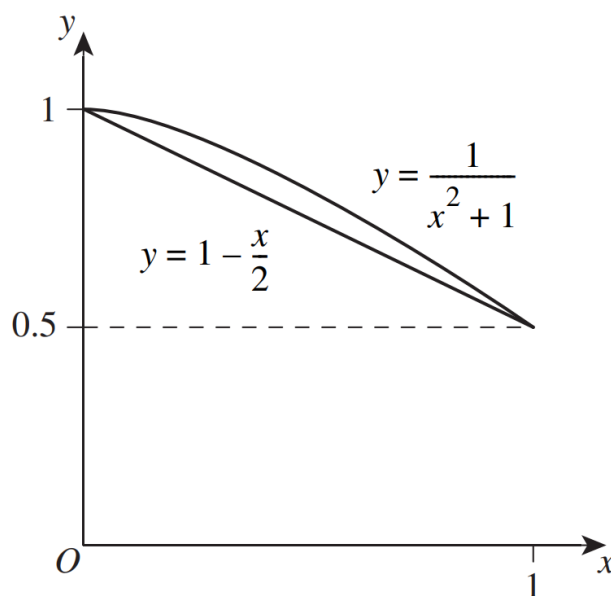
(You may use the diagram below to help solve the problem)



4. (Sydney Grammar 2023 Q14a) A container is formed by rotating the curve $y = 10 \tan^{-1} \frac{x}{4}$ around the y -axis from $y = 0$ cm to $y = 12$ cm as shown in the diagram. The container is filled to the top with water.



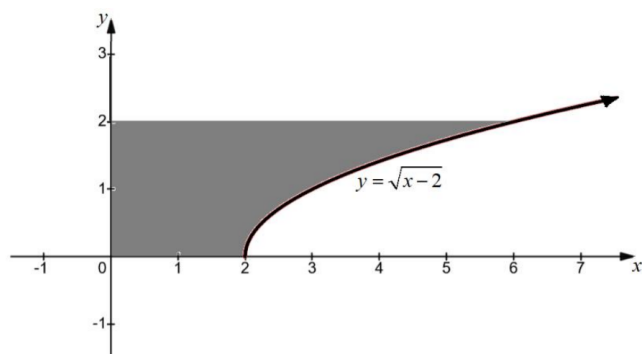
- (i) By noting that $\tan^2 \theta = \sec^2 \theta - 1$, show that the initial volume V_0 of water in the container is 690 mL, to the nearest mL. **3**
5. (Riverview 2020 Q12c) The diagram shows the graph of $y = \frac{1}{x^2+1}$ and the graph of $y = 1 - \frac{x}{2}$ for $0 \leq x \leq 1$.
- (i) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = \frac{1}{x^2+1}$, the y -axis and the line $y = \frac{1}{2}$ is rotated 360° about the y -axis. **2**
- (ii) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = 1 - \frac{x}{2}$, the y -axis and the line $y = \frac{1}{2}$ is rotated 360° about the y -axis. **2**
- (iii) Use the results from parts (c) (i) and (ii) to show that $\ln 2 > \frac{2}{3}$. **1**



6. (Caringbah 2021 Q12c) The graph above is the function $f(x) = 2 \cos^{-1} \frac{x}{3}$.



- (i) Find the value of the y intercept. **1**
 - (ii) Find the domain and range of $f(x) = 2 \cos^{-1} \frac{x}{3}$. **2**
 - (iii) Calculate the area of the region under the function $f(x) = 2 \cos^{-1} \frac{x}{3}$ between $x = 0$ and $x = 3$. **2**
7. (Penrith 2025 Q11a) The shaded region bounded by the curve $y = \sqrt{x-2}$, the coordinate axes and the line $y = 2$ shown below is rotated about the y -axis. Calculate the volume of the solid generated. **3**



8. (Penrith 2025 Q12c) A solid of revolution is to be found by rotating the region bounded by the x -axis and the curve $y = (a+2)\sin(ax)$, where $a > 0$, between $x = 0$ and $x = \frac{\pi}{2a}$ about the x -axis. See the diagram shown. Find the value of a for which the volume is $2\pi^2$ cubic units. **3**

